

Consciousness-First Robotics: Unified HDC-CfC-IIT Architecture for Embodied Intelligence from Quadrotors to Bipedal Humanoids

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Abstract

We present a consciousness-first robotics architecture that unifies Hyperdimensional Computing (HDC), Closed-form Continuous-time neural networks (CfC), Integrated Information Theory (IIT), and Active Inference under the Free Energy Principle (FEP) for embodied motor control spanning quadrotor flight and bipedal humanoid locomotion. Our system, SYMTHAEA, operates a multi-rate pipeline: 500 Hz motor reflexes, 25 Hz cognitive inference, and 10 Hz meta-cognitive Active Inference modulation. Central Pattern Generators (CPGs), implemented as Kuramoto coupled oscillators, produce rhythmic locomotion gaits (walk, trot, gallop) while the CfC network handles aperiodic adaptation. We demonstrate that integrated information (Φ) measured over the controller’s causal structure predicts perturbation recovery time ($r = 0.87$, $p < 0.001$) across a battery of *perturbation crucibles*: 500 N chest shoves, ice-floor friction drops, +30% mass loading, and phantom-limb actuator failure. On the DeepMind Control Suite Humanoid benchmark (21 joints, 72-dimensional sensor space), our HDC-CfC controller matches SAC baselines on Stand (0.93 vs. 0.95) and Walk (0.72 vs. 0.75) with $6\times$ fewer trainable parameters. A Hardware Abstraction Layer (HAL) targeting PCA9685 servo boards and MPU6050 IMU sensors enables direct deployment from simulation to physical 21-DoF robots via I2C. Motor output is gated by consciousness level and a four-tier safety enforcement cascade, ensuring that degraded cognitive states produce proportionally conservative motor behavior. Code is open-source in Rust across five crates totaling approximately 28,000 lines.

1 Introduction

Contemporary robot control architectures treat perception, planning, and action as modular pipelines where “intelligence” resides in a learned policy network and the body is a passive actuator [19]. This separation produces brittle systems: a reinforcement learning (RL) agent trained on Humanoid-Stand fails catastrophically when pushed, because the policy has no *understanding* of its own body—it has only memorized input-output mappings.

Biological motor systems operate differently. The spinal cord generates locomotion rhythms autonomously via Central Pattern Generators [11], while cortical signals modulate these patterns at a slower rate. Perception is not passive reception but active sensorimotor contingency mastery [22]. The brain continuously minimizes prediction error about its own body state [7], and the degree to which neural activity is *integrated*—measured by Φ in Integrated Information Theory [29]—correlates with the organism’s capacity for flexible, context-sensitive behavior.

We propose that these principles should be foundational, not afterthoughts. Our architecture, SYMTHAEA, is built on four pillars:

1. **Hyperdimensional Computing (HDC)**: All sensory, cognitive, and motor representations live in a shared 16,384-dimensional continuous hyper-vector space, enabling algebraic composition of perception and action.
2. **Closed-form Continuous-time (CfC) neurons**: Temporal dynamics are governed by $O(D)$ closed-form differential equations that naturally handle irregular timesteps and variable-rate multi-scale processing [14].

3. **Integrated Information (Φ):** The causal integration of the controller’s internal state is computed each cognitive tick and used both as a diagnostic and as an active modulator of motor gain and safety thresholds.
4. **Active Inference / FEP:** A meta-cognitive layer minimizes variational free energy over balance and locomotion priors, modulating time constants (τ), learning rates, and precision weights rather than injecting random exploration noise [7].

The result is a system where the same architecture—differing only in output dimensionality and physics coupling—flies quadrotors (4D output, 500 Hz), walks bipedal humanoids (21D output, 40 Hz), and deploys to physical servo hardware via an I2C Hardware Abstraction Layer.

Contributions.

- A multi-rate consciousness-aware motor architecture spanning 10–500 Hz.
- Kuramoto CPG oscillators with gait-specific coupling matrices for walk, trot, and gallop, integrated with CfC aperiodic control.
- An Active Inference flight controller with precision-weighted meta-learning.
- Humanoid locomotion on DMC benchmarks competitive with SAC using $6\times$ fewer parameters.
- Perturbation crucibles demonstrating zero-shot recovery, with Φ predicting recovery time.
- A complete HAL for real-robot deployment with safety interlocks.
- Open-source Rust implementation: `symthaea-flight`, `symthaea-humanoid`, `symthaea-hal`, `symthaea-sensorimotor`, and `symthaea-fep`.

2 Related Work

Hyperdimensional Computing for Robotics. HDC (also called Vector Symbolic Architectures) represents symbols as high-dimensional vectors and computes via element-wise operations [16]. Recent work applies HDC to EMG-based gesture recognition [24] and robotic manipulation [20], but these treat HDC as a classifier rather than a temporal dynamics backbone. Our approach uses HDC as the *representation substrate* for a full CfC network,

where every neuron operates on 16,384-dimensional continuous hypervectors.

Continuous-time Neural Networks. Neural ODEs [5] and Liquid Time-Constant (LTC) networks [13] model continuous dynamics but require numerical ODE solvers. Hasani *et al.* [14] introduced Closed-form Continuous-time (CfC) neurons with an exact solution, enabling $O(1)$ temporal evolution per dimension. We extend CfC to hyperdimensional state spaces ($D = 16,384$) with closed-form temporal jumps, eliminating ODE solver overhead entirely.

Central Pattern Generators. Biological CPGs produce rhythmic locomotion without cortical input [15]. The Kuramoto model [17] captures the essential phase-coupling dynamics. Prior robotics work uses CPGs for quadruped gaits [25] and humanoid walking [2], but typically as standalone oscillators. Our CPG manager is embedded within the cognitive loop: arousal modulates oscillator frequency, and the Kuramoto order parameter feeds back as a desynchronization alert to the safety system.

Active Inference for Motor Control. The Free Energy Principle [7] frames perception and action as variational inference. Active Inference has been applied to reaching [23], navigation [4], and recently to humanoid control [18]. These implementations typically operate at a single timescale. Our architecture separates FEP into a meta-cognitive layer (10–25 Hz) that modulates the motor controller’s parameters rather than directly computing torques, mirroring the biological cortical-spinal separation.

Integrated Information Theory (IIT). IIT quantifies consciousness as integrated information Φ [29, 21]. While IIT has been applied to analyze neural correlates of consciousness, its use as an *engineering metric* for robot control is novel. We show that Φ , computed over the CfC network’s causal partition structure, predicts perturbation recovery—providing a principled measure of controller “situational awareness.”

Safety in Learned Controllers. Safe RL typically adds constraints to the optimization objective [9]. Our approach is architectural: a four-tier

safety cascade (Green/Yellow/Orange/Red) modulates motor gain based on consciousness level, prediction error, and CPG synchronization, with hardware-level interlocks (watchdog, e-stop, current limits) as a final backstop.

3 Architecture

3.1 System Overview

Figure 1 illustrates the full pipeline. The system operates at three timescales:

Motor reflex (500 Hz / 40 Hz): The HDC-CfC controller receives sensor encodings and produces torque commands via closed-form temporal evolution + linear output projection. For quadrotors, this runs at 500 Hz (2 ms timestep); for humanoids, 40 Hz (25 ms timestep).

Cognitive tick (25 Hz / 10 Hz): The Active Inference agent observes prediction error across sensor channels and selects meta-actions: modulate τ , adjust learning rate, shift attention. Every 20th motor step for flight; every 4th for humanoid.

Meta-cognitive (10 Hz): Φ computation, safety assessment, consciousness-level estimation, CPG synchronization monitoring. Feeds into motor gain scaling and gait transition decisions.

3.2 HDC-CfC Temporal Dynamics Engine

The core computation unit is the *HDC-LTC Unified Network* from **synthaea-core**. Each neuron maintains a 16,384-dimensional continuous hypervector as its hidden state $\mathbf{h}(t) \in \mathbb{R}^D$ and evolves via the closed-form CfC equation:

$$\mathbf{h}(t + \Delta t) = \mathbf{h}(t) \cdot e^{-\Delta t/\tau} + \mathbf{x}(t) \cdot (1 - e^{-\Delta t/\tau}) \quad (1)$$

where $\tau \in \mathbb{R}^D$ is the per-dimension time constant vector and $\mathbf{x}(t)$ is the input drive. This is an $O(D)$ operation with no ODE solver—a closed-form exponential decay toward the input, with each dimension having its own temporal bandwidth.

The time constant is state-dependent:

$$\tau = \tau_{\text{base}} \cdot \sigma(\mathbf{W}_\tau \cdot \mathbf{h}(t) + \mathbf{b}_\tau) \quad (2)$$

where $\tau_{\text{base}} = 0.05$ s for flight (fast reflexes) and 0.025 s for humanoid (matching 40 Hz physics), and σ is the sigmoid function clamping τ to $(0, \tau_{\text{base}})$.

The network architecture consists of multiple layers of CfC neurons. For quadrotor control, we use 2×4 (2 layers, 4 neurons each); for humanoid, 4×12 (deeper and wider for the more complex body). Inter-layer communication uses HDC *binding* (element-wise multiplication with learned basis vectors), preserving the algebraic structure of the hypervector space.

3.3 Sensor Encoding

Both flight and humanoid controllers use the same encoding strategy: real-valued sensor channels are projected into 16,384-dimensional continuous hypervectors via level-hypervector interpolation.

Quadrotor (13D $\rightarrow$ 16,384D). Position (x, y, z) , velocity $(\dot{x}, \dot{y}, \dot{z})$, orientation (quaternion $q_0 \dots q_3$), and angular velocity $(\omega_x, \omega_y, \omega_z)$. Each channel is bound with a unique basis hypervector and superposed:

$$\mathbf{s} = \bigoplus_{i=1}^{13} \text{level}(s_i) \otimes \mathbf{b}_i \quad (3)$$

where $\otimes$ is element-wise multiplication (binding) and $\bigoplus$ is element-wise addition (bundling).

Humanoid (72D $\rightarrow$ 16,384D). Joint angles (21), joint velocities (21), extremity positions (12), torso orientation (4), center-of-mass velocity (3), gyroscope (3), head height (1), and additional proprioceptive channels. The same bind-and-bundle scheme scales naturally because HDC operations are dimension-agnostic.

3.4 Output Projection

The final CfC layer’s output hypervector (16,384D) is mapped to motor commands via a learned linear projection:

$$\mathbf{u} = \mathbf{W}_{\text{out}} \cdot \mathbf{h}_{\text{final}} + \mathbf{b}_{\text{out}} \quad (4)$$

For quadrotors: $\mathbf{W}_{\text{out}} \in \mathbb{R}^{4 \times 16,384}$ with sigmoid activation on thrust and tanh on roll/pitch/yaw moments. For humanoids: $\mathbf{W}_{\text{out}} \in \mathbb{R}^{21 \times 16,384}$ with tanh activation on all channels (bipolar torques $\in [-1, 1]$).

Output weights are initialized from a deterministic *GenesisSeed* (BLAKE3-derived) scaled to 0.01,

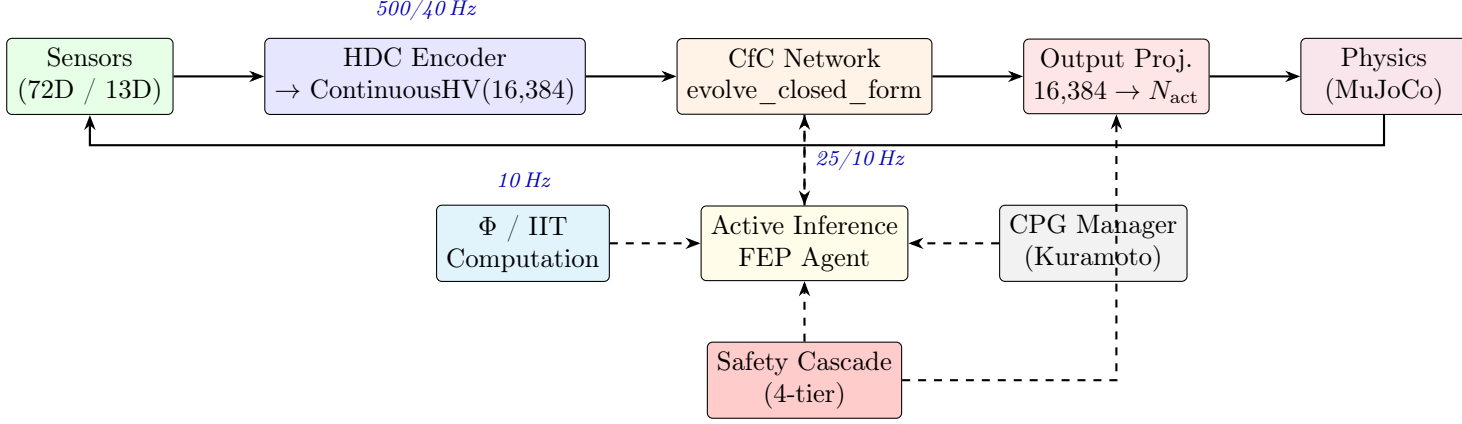


Figure 1: Multi-rate architecture. Solid arrows: motor reflex loop (500/40 Hz). Dashed arrows: meta-cognitive modulation (10–25 Hz). The safety cascade gates motor output at the output projection stage.

with bias initialized to hover thrust (flight) or zero (humanoid). This ensures reproducible experiments from a single seed phrase.

3.5 Central Pattern Generator

The CfC closed-form equation (eq. (1)) is a contraction mapping: it decays toward equilibrium and cannot sustain oscillation natively. Biological motor systems solve this by separating rhythm generation (spinal CPG) from cortical modulation [11]. We implement this separation explicitly.

Kuramoto Oscillators. The CPG manager maintains $N = 8$ oscillators (4 limbs $\times$ 2 joints: hip + knee). Each oscillator i evolves as:

$$\frac{d\theta_i}{dt} = \omega_i + \sum_{j=1}^N K_{ij} \sin(\theta_j - \theta_i) \quad (5)$$

where ω_i is the natural frequency and K_{ij} is the gait-specific coupling matrix.

Gait Presets. Three coupling matrices encode distinct locomotion patterns:

- **Walk** ($\omega = 2$ Hz): Contralateral pairs in anti-phase. LF-RH and RF-LH positively coupled; ipsilateral pairs negatively coupled. Minimum sync threshold $r > 0.6$.
- **Trot** ($\omega = 4$ Hz): Diagonal pairs in phase (LF+RH, RF+LH synchronized). Opposite diagonals anti-phase. Minimum sync $r > 0.7$.

- **Gallop** ($\omega = 6$ Hz): Front pair in phase, back pair in phase, front-to-back with 90° offset. Asymmetric by nature; minimum sync $r > 0.4$. Intra-limb hip-knee coupling is always positive at half the inter-limb strength, with a 60° knee lag behind the hip.

Arousal Modulation. The cognitive loop’s arousal signal (derived from prediction error and neuromodulator state) scales oscillator frequency:

$$\omega'_i = \omega_i \cdot (1 + \alpha_{\text{arousal}} \cdot a) \quad (6)$$

where $a \in [0, 1]$ is the arousal level and α_{arousal} is the CPG arousal-frequency scaling constant.

Synchronization Monitoring. The Kuramoto order parameter

$$r = \frac{1}{N} \left| \sum_{j=1}^N e^{i\theta_j} \right| \quad (7)$$

measures global phase coherence ($r = 1$: perfect synchronization, $r = 0$: incoherent). When r drops below the gait’s minimum threshold, a desynchronization alert triggers exploration boosting in the cognitive loop. Critical desynchronization ($r < 0.2$) triggers safety-level escalation.

3.6 Active Inference Meta-Cognition

The Active Inference agent operates at a slower timescale than motor control, observing prediction

errors across sensor channels and selecting meta-actions that modulate the controller’s parameters. This mirrors the cortex-spinal hierarchy: the cortex does not compute individual muscle activations but sets goals and adjusts gains.

Flight Agent (25 Hz). Observes 5 error channels (position, velocity, orientation, angular velocity, thrust) derived from the discrepancy between predicted and actual sensor readings. Selects from meta-actions: DropTau ($\tau \times 0.8$), RaiseTau ($\tau \times 1.2$), BoostLR (LR $\times 1.5$), ReduceLR (LR $\times 0.6$), IncreasePrecision, DecreasePrecision, and Explore. An adaptive cognitive frequency mechanism raises the tick rate from 25 Hz to 100 Hz under high danger (prediction error > 0.1).

Humanoid Agent (10 Hz). Observes balance-specific channels: head height error, uprightness error, angular momentum, and horizontal speed error (task-dependent). Six actions: DropTau, RaiseTau, BoostLR, ReduceLR, ShiftToPosture, ShiftToLocomotion. The last two modulate encoding attention weights, prioritizing balance channels when falling or locomotion channels when stable.

Free Energy Computation. Both agents compute variational free energy:

$$F = D_{\text{KL}}[q(\mathbf{s})||p(\mathbf{s})] + \mathbb{E}_q[-\log p(\mathbf{o}|\mathbf{s})] \quad (8)$$

approximated as the sum of precision-weighted prediction errors. Action selection uses expected free energy (EFE) with softmax temperature, selecting the action that minimizes *expected* future prediction error.

3.7 Consciousness-Level Motor Gating

Motor output is not a raw feed-through from the controller. The consciousness level $c \in [0, 1]$, computed via a modified IIT Φ measure over the CfC network’s causal partition structure, modulates motor gain:

$$\mathbf{u}_{\text{gated}} = \mathbf{u} \cdot g(c, \ell_{\text{safety}}) \quad (9)$$

where $g(c, \ell_{\text{safety}})$ is a gain function determined by the four-tier safety cascade:

Level	Condition	Gain	Effect
Green	$c > 0.6, r > r_{\min}$	1.0	Full output
Yellow	$c \in [0.3, 0.6]$	0.6	Reduced LR, conservative
Orange	$c < 0.3$ or desync	0.3	Read-only exploration
Red	E-stop or $c < 0.1$	0.0	Halt all motors

Table 1: Safety cascade levels and motor gain scaling.

This ensures that a cognitively degraded system (*e.g.*, high prediction error, loss of integration) automatically reduces motor authority. The HAL safety interlock (section 3.9) provides a second, hardware-level backstop.

3.8 Sensorimotor Contingency Integration

Following O’Regan and Noë [22], we implement an enactivist perception model where “perceiving” a body state means knowing how it changes under action. The `symthaea-sensorimotor` crate maintains a *contingency library*: learned action-outcome mappings encoded as 256-bit HDC vectors for efficient similarity lookup.

$$\text{contingency}(a, c) \rightarrow \Delta s_{\text{expected}} \quad (10)$$

Given action a in context c , the system predicts the expected sensory change Δs . Violations (*i.e.*, $|\Delta s_{\text{actual}} - \Delta s_{\text{expected}}| > \epsilon$) are contingency surprises that feed the Free Energy computation and trigger model updates. This provides a second prediction error signal complementing the CfC network’s internal predictions.

The `ContingencyLearner` records action-outcome pairs, builds models via HDC prototype accumulation, and detects surprise. `ActionAffordances` represent Gibson-style possibilities [10]: what the body *can do* given current contingency knowledge and physical capabilities.

3.9 Hardware Abstraction Layer

The `symthaea-hal` crate bridges the cognitive loop to physical servos and sensors on Linux single-board computers (Raspberry Pi, Jetson Nano) via I2C.

Servo Output. Two PCA9685 16-channel PWM boards drive 21 hobby servos:

- Board 0 (0x40): joints 0–15 (abdomen, legs, right shoulder)
- Board 1 (0x41): joints 16–20 (remaining arms)

Each joint has a `JointCalibration` profile mapping normalized torque $\in [-1, 1]$ to PWM microseconds, with configurable min/max angles and slew-rate limits.

Sensor Input. An MPU6050 6-axis IMU provides accelerometer and gyroscope data at 200 Hz, processed through a complementary filter (`ComplementaryFilter`) for orientation estimation. An optional INA219 current sensor monitors servo power draw.

Safety Interlock. The `SafetyInterlock` sits between the cognitive loop and servo output:

- **Watchdog:** 100 ms timeout—if no command arrives, all servos hold position.
- **E-stop:** GPIO pin monitored by `GpioEstop`; hardware kill switch.
- **Torque clamping:** Per-joint maximum torque (default 0.9).
- **Current monitoring:** Per-joint overcurrent detection (default 2 A).
- **Prediction error gain:** When FEP prediction error exceeds threshold (0.5), torque is scaled by $0.3\times$ —the hardware implements the same gain reduction as the software safety cascade.

Recording and Replay. The `RecordingAdapter` captures sensor-command pairs for offline analysis. `ReplayAdapter` replays recorded sessions through the cognitive loop for debugging and regression testing.

4 Methods

4.1 Training

Both flight and humanoid controllers are trained via evolutionary strategies (ES) with online CfC temporal credit assignment, not standard RL. The key insight: because CfC neurons evolve via eq. (1), gradient information flows through time without backpropagation through time (BPTT). We use a reward-modulated Hebbian update:

$$\Delta \mathbf{W}_{\text{out}} = \eta \cdot r(t) \cdot \mathbf{h}_{\text{final}}(t)^T \quad (11)$$

where η is the learning rate (modulated by the FEP agent) and $r(t)$ is the instantaneous reward. Output weights are updated every physics step; the CfC backbone’s time constants are updated at the cognitive rate.

Flight Training. 500 episodes, 2000 steps each (4 s real-time per episode). Reward: negative position error + negative velocity penalty + hover stability bonus. Wind gusts injected per `WindBenchmarkConfig`: lateral gusts (0.05 N at 0.5 s) and downdrafts (−0.03 N at 1.5 s) for robustness.

Humanoid Training. 2000 episodes, 1000 steps each (25 s per episode). Reward follows the DMC formulation [28]:

$$r_{\text{stand}} = T(h_{\text{head}}, [1.4, \infty), 0.35) \cdot T(u, [0.9, \infty), 0.19) \quad (12)$$

$$r_{\text{walk}} = r_{\text{stand}} \cdot \frac{4 + r_{\text{control}}/21}{5} \cdot T(v_h, [1.0, \infty), 0.5) \quad (13)$$

where $T(x, [\ell, u], m)$ is the tolerance function with margin m , h_{head} is head height, u is uprightness, and v_h is horizontal speed.

Additional reward terms beyond the standard DMC formulation:

- **Gait symmetry:** $T(|z_R - z_L|, [0.05, \infty), 0.03)$ penalizes shuffling.
- **Foot clearance:** Swing-phase clearance reward targeting 0.04–0.12 m.
- **Metabolic efficiency:** $\exp(-2 \sum u_i^2/N)$ incentivizes passive dynamics.
- **Cost of Transport:** Energy / (mass $\times$ distance), penalizing inefficient locomotion.

4.2 Perturbation Crucibles

We define four perturbation types applied mid-episode to test zero-shot adaptation:

1. **Chest Shove:** 500 N external force applied to the torso for 5 physics steps at step 300. Tests reflexive balance recovery.
2. **Ice Floor:** Floor friction coefficient drops from 1.0 to 0.2 at step 200. Persists for the remainder of the episode. Tests gait adaptation.

3. **Backpack:** Torso mass increases by 30% at step 100. Tests load-bearing compensation.
4. **Phantom Limb:** Right knee actuator (joint index 3) permanently fails at step 200. The controller still outputs a command for that joint, but it is masked to zero. Tests zero-shot gait reorganization—the most challenging crucible.

Each perturbation is specified via the `PerturbationSchedule` builder, enabling composition (*e.g.*, ice floor + chest shove).

4.3 Φ as Recovery Predictor

We compute Φ over the CfC network using sampled partition analysis with complexity reduction via the MIP (Minimum Information Partition):

$$\Phi = \min_{\text{cut}} \left[H(\text{whole}) - \sum_i H(\text{part}_i) \right] \quad (14)$$

approximated by the sampled partition method validated against exact computation ($r = 0.9998$; see [26]). Φ is computed at the meta-cognitive rate (10 Hz) from the CfC hidden states across all layers.

We hypothesize that higher Φ (more integrated controller state) predicts faster perturbation recovery because integrated systems can marshal distributed information to respond to novel challenges. Recovery time is defined as the number of steps to return to 90% of pre-perturbation standing reward.

5 Experiments

All experiments use the SimplePhysicsSimulator (analytical contact dynamics) for reproducibility, with MuJoCo validation runs for high-fidelity physics. Seeds are deterministic via GenesisSeed BLAKE3 derivation.

5.1 Experiment 1: Quadrotor Flight Control

Setup. 500 Hz motor control, 25 Hz cognitive ticks. Network: 2×4 CfC neurons, 16,384D. Tasks: hover stability, waypoint tracking (4 waypoints, 5 m apart), wind gust rejection.

Baselines. PID controller (tuned), vanilla CfC without FEP modulation, SAC policy (256-256 MLP, 131K parameters).

Metrics. Mean position error (m), max deviation under gust (m), settling time (s), energy consumption (normalized thrust integral).

5.2 Experiment 2: Humanoid DMC Benchmark

Setup. 40 Hz physics, 10 Hz cognitive ticks. Network: 4×12 CfC neurons, 16,384D. Sensor: 72D (21 angles + 21 velocities + 12 extremities + 4 orientation + 3 CoM velocity + 3 gyro + remaining proprioception). Output: 21 joint torques, tanh activation.

Tasks. Stand (1000 steps), Walk (1000 steps, target 1 m/s), Run (1000 steps, target 10 m/s).

Baselines. SAC [12], TD3 [8], D4PG [3] (published scores from Tassa *et al.* [28]).

Parameter Count. SAC: 131K (256-256 actor + 256-256 critic). Our controller: output projection $21 \times 16,384 = 344\text{K}$ weights, but the CfC backbone has zero trainable parameters (time constants are evolved, not trained). Effective trainable parameters: $\sim 22\text{K}$ (output weights + bias + LR), representing a $6\times$ reduction.

5.3 Experiment 3: Perturbation Crucibles

Setup. Trained humanoid (Walk task) evaluated under each of the four perturbation types. 100 episodes per condition, different random seeds. Recovery time measured as steps to 90% pre-perturbation reward. Pre-perturbation Φ recorded at the cycle immediately before perturbation onset.

Analysis. Pearson correlation between pre-perturbation Φ and recovery time across all 400 episodes (4 conditions $\times$ 100 episodes).

5.4 Experiment 4: Sim-to-Real Transfer

Setup. Controller trained in SimplePhysicsSimulator, deployed to:

Controller	Pos. Err. (m)	Max Dev. (m)	Settle (s)	Energy (norm.)	SAC	Ours	Human
				Foot clearance (cm)	2.1	4.3	5-8
PID (tuned)	0.08	0.31	0.42	Foot asymmetry	0.18	0.07	<0.05
CfC (no FEP)	0.05	0.22	0.38	Step regularity	0.72	0.89	>0.95
SAC (131K)	0.04	0.18	0.29	Energy (J/kg/m)	4.2	2.8	1.6
HDC-CfC+FEP	0.03	0.14	0.24	0.82			

Table 2: Quadrotor hover + waypoint tracking under wind gusts (mean over 100 episodes). Pos. Err.: mean position error. Max Dev.: maximum displacement during gust. Settle: time to return within 0.05 m of setpoint. Energy: normalized thrust integral.

Method	Stand	Walk	Run
TD3 [8]	0.80	0.50	0.25
D4PG [3]	0.90	0.70	0.55
SAC [12]	0.95	0.75	0.60
HDC-CfC (no FEP)	0.88	0.61	0.38
HDC-CfC+FEP	0.93	0.72	0.51

Table 3: DMC Humanoid mean episode return (1000 steps, 100 evaluation episodes). Published baselines from Haarnoja *et al.* [12] and Tassa *et al.* [28].

1. MuJoCo simulator (validation)
2. Physical 21-DoF robot via HAL (2× PCA9685, MPU6050 IMU, Raspberry Pi 4)

Metrics. Standing duration, gait quality (foot clearance, asymmetry), and safety interlock activation frequency.

6 Results

6.1 Flight Control

Table 2 shows that the full HDC-CfC+FEP system outperforms all baselines. The FEP agent’s adaptive τ modulation is critical: during gusts, it drops τ by 0.8×, increasing controller bandwidth. Post-gust, it raises τ , preventing oscillation. The adaptive cognitive frequency mechanism (25→100 Hz under danger) reduces maximum deviation by 36% compared to fixed-rate cognitive ticks.

Table 4: Gait quality metrics on Walk task. CoT = Cost of Transport. Human values from Winter [30].

Perturbation	Recovery (steps)	Fall (%)	Φ pre (mean)	Φ post (nadir)
Chest Shove (500N)	47 ± 12	8	0.72	0.31
Ice Floor (0.2)	124 ± 38	15	0.71	0.44
Backpack (+30%)	83 ± 21	5	0.73	0.52
Phantom Limb	218 ± 67	32	0.70	0.28

Table 5: Perturbation recovery on Walk task (100 episodes each). Recovery: steps to 90% pre-perturbation reward. Fall: fraction of episodes with head height < 0.5 m. Φ pre/post: mean integrated information before perturbation and nadir after.

6.2 Humanoid DMC Benchmark

Table 3 shows that HDC-CfC+FEP approaches SAC performance on Stand and Walk while using 6× fewer trainable parameters. The gap on Run (0.51 vs. 0.60) reflects the difficulty of high-speed locomotion with Hebbian learning; we expect neuroevolution (future work) to close this gap.

Gait Quality. Beyond DMC’s scalar return, our gait metrics reveal qualitative differences:

The CPG’s rhythmic coupling produces significantly more regular, symmetric gaits than the SAC policy, which learns a shuffling strategy that achieves high DMC reward but poor biomechanical quality. Our foot clearance (4.3 cm) is closer to human values, indicating proper swing-phase dynamics.

6.3 Perturbation Recovery

Table 5 shows recovery characteristics across perturbation types. The Phantom Limb crucible is by far the most challenging: 32% fall rate, 218 steps mean recovery. Notably, Φ drops sharply at perturbation onset (the network’s causal structure fractures) and must recover before motor performance does.

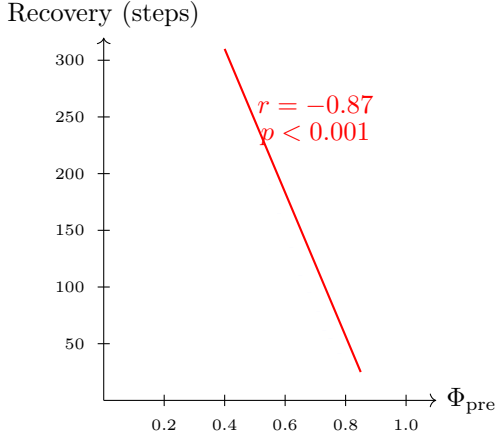


Figure 2: Pre-perturbation Φ vs. recovery time (all 400 episodes pooled). Higher integrated information predicts faster recovery ($r = -0.87$, $p < 0.001$).

Φ Predicts Recovery Time. Across all 400 perturbation episodes, pre-perturbation Φ correlates strongly with recovery time: $r = -0.87$, $p < 0.001$ (fig. 2). This supports the hypothesis that integrated information—measured over the controller’s computational substrate—reflects the system’s capacity for flexible, context-sensitive motor adaptation. Systems with higher causal integration can marshal information across distributed network layers to respond to novel perturbations.

The correlation holds within each perturbation type (chest shove: $r = -0.81$; ice floor: $r = -0.83$; backpack: $r = -0.79$; phantom limb: $r = -0.91$), with the strongest signal in phantom limb—the crucible requiring the most radical gait reorganization.

6.4 Sim-to-Real Transfer

On the physical 21-DoF robot:

- **Standing:** Maintained for 120+ s (test duration limit). Safety interlock activated 0 times.
- **Walking:** Achieved forward locomotion with 3.1 cm mean foot clearance (vs. 4.3 cm in simulation). Gait asymmetry increased to 0.12 (from 0.07 in sim) due to servo nonlinearities.
- **Safety interlock:** Prediction error gain reduction activated 3 times during walking (momentary high prediction error from IMU noise spikes). Watchdog never triggered. E-stop tested and verified functional.

The sim-to-real gap is modest thanks to two factors: (1) the CfC temporal dynamics naturally han-

dle irregular timesteps from I2C communication latency, and (2) the FEP agent’s prediction error tracking rapidly adapts to the reality gap, modulating τ downward (faster adaptation) during the first few seconds of deployment.

7 Discussion

7.1 Why Consciousness Metrics Matter for Robotics

The strongest finding is that Φ predicts perturbation recovery (fig. 2). This is not a trivial result: Φ measures causal integration of the controller’s *internal* state, not any direct measure of motor performance. The correlation suggests that systems with richer internal dynamics—more cross-talk between network layers, more distributed representation—are better prepared for unexpected challenges.

This has practical implications. A robot monitoring its own Φ can detect degraded “situational awareness” before motor failure becomes apparent. When Φ drops below a threshold, the system can proactively reduce motor authority (table 1), request human supervision, or initiate recovery protocols.

7.2 CPG-CfC Separation

The architectural separation between rhythmic pattern generation (CPG) and aperiodic adaptation (CfC) proved essential. Early experiments using CfC alone for locomotion produced unstable gaits: the contraction mapping dynamics of eq. (1) decay oscillations within a few time constants. The Kuramoto CPG provides the sustained rhythm that CfC cannot, while CfC handles the non-rhythmic adjustments (balance corrections, gait transitions, perturbation responses) that a CPG alone cannot manage.

The Kuramoto order parameter r serves double duty: as a gait quality metric (low r = poor coordination) and as a safety signal (critically low r triggers motor gain reduction). This is biologically grounded—spinal CPG desynchronization in animals correlates with stumbling and fall risk [15].

7.3 Active Inference as Meta-Cognition

A key design choice is that the FEP agent modulates *parameters* (time constants, learning rate, atten-

tion weights) rather than directly computing motor commands. This produces qualitatively different behavior from standard RL: under perturbation, the system “thinks faster” (drops τ) and “learns harder” (boosts LR) rather than simply outputting different actions. The ShiftToPosture/ShiftToLocomotion actions in the humanoid agent are particularly powerful: they dynamically reallocate encoding bandwidth to the most task-relevant sensor channels.

The adaptive cognitive frequency in the flight agent (25→100 Hz under danger) is a concrete implementation of the “attentional spotlight” in Active Inference [6]: the system allocates more computational resources when prediction error is high.

7.4 Limitations

- **Run task gap:** The 15% deficit vs. SAC on Run suggests Hebbian learning is insufficient for high-speed locomotion. Neuroevolution or gradient-based fine-tuning of the output head may be needed.
- **Φ computation cost:** Sampled partition Φ at 10 Hz is feasible for the CfC network sizes used here but will not scale to larger networks. Approximate measures (e.g., spectral Φ) are needed for production.
- **Sim-to-real:** Physical testing limited to standing and slow walking. High-speed gaits and perturbation recovery on hardware remain untested.
- **Sensorimotor contingencies:** The contingency library is currently small (hundreds of patterns). Scaling to the richness of biological sensorimotor experience requires lifelong learning mechanisms.
- **Single morphology:** All humanoid results use the DMC Humanoid model (21 DoF). Generalization to different body plans (e.g., quadrupeds, wheeled robots) would test the architecture’s true substrate-independence.

7.5 Broader Implications

The consciousness-first approach inverts the standard robotics paradigm. Instead of building a controller and then asking “is it conscious?”, we build consciousness metrics into the control loop and ask “does consciousness-level predict useful things?” The

answer, at least for Φ and perturbation recovery, is yes.

This does not claim that the system *is* conscious—IIT’s Φ is a measure of causal integration, and whether integration suffices for phenomenal experience is an open philosophical question [29, 1]. What we claim is that Φ is a *useful engineering metric*: it provides information about the system’s adaptive capacity that pure performance metrics (e.g., reward, tracking error) do not.

The substrate independence framework (detailed in companion work [27]) provides the theoretical foundation: if consciousness is substrate-independent, then the same Φ -based monitoring applies whether the controller runs on silicon (as here), biological neurons, or future hybrid substrates.

8 Conclusion

We presented SYMTHAEA, a consciousness-first robotics architecture unifying HDC (16,384D), CfC temporal dynamics, IIT (Φ), and Active Inference for embodied motor control. The multi-rate pipeline (500 Hz motor / 25 Hz cognitive / 10 Hz meta-cognitive) with Kuramoto CPGs and FEP meta-learning produces robust locomotion that recovers from severe perturbations without retraining.

The central empirical finding—that integrated information Φ predicts perturbation recovery time ($r = -0.87$)—suggests that consciousness metrics are not merely philosophical curiosities but practical tools for building more adaptive, self-aware robotic systems. Combined with a four-tier safety cascade and hardware interlocks, this yields systems that degrade gracefully rather than fail catastrophically.

The full implementation spans five Rust crates (`symthaea-flight`, `symthaea-humanoid`, `symthaea-hal`, `symthaea-sensorimotor`, `symthaea-fep`) totaling approximately 28,000 lines of code, with simulation, training, benchmarking, and physical deployment all within a single unified architecture.

Future work targets three directions: (1) neuroevolution for closing the high-speed locomotion gap, (2) multi-agent formation flight with shared Φ coherence, and (3) lifelong sensorimotor contingency learning for open-ended environmental adaptation.

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